

Shapes of Geometric Consciousness: Van Hiele’s Levels, Hegel’s *Phenomenology*, and the Recognitive Engine of Inquiry-Based Geometry for Preservice Teachers

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Abstract

This paper argues for a partial structural alignment between Hegel’s “shapes of consciousness” (*Gestalten des Bewusstseins*) in the *Phenomenology of Spirit* and the van Hiele levels of geometric thought, and uses the alignment to specify the recognitive mechanism that drives transitions between van Hiele levels in inquiry-based geometry courses for preservice teachers. The mathematics-education literature has not made this alignment directly. The closest extant work uses Hegel via Lakatos’s dialectic (Ernest, 2016), via Radford’s objectification framework in algebra (Radford, 2008), or via Activity Theory readings of workplace mathematics (LaCroix, 2014); it uses Brandom and Habermas in mathematics-education contexts (Bakker et al., 2015; Sensevy et al., 2013; Rüede, 2013; Ongstad, 2006; Díez-Palomar & Olivé, 2015) but not in connection with van Hiele transitions; and it identifies “social climates” (Bowers, Brandt, Stovall, & Vargas, in JRME) and “explorative discourse” (Mahlaba & Mahlobo, 2022) as the mechanism for level transitions without naming the recognitive structure that explains why discourse is the mechanism. We argue that the structure is recognitive in the precise sense developed by Brandom (1994, 2019) and refined by Carspecken in his synthesis of Brandom and Habermas. Brandom’s “trust” (the moment-by-moment dyadic scorekeeping of commitments and entitlements) and Habermas’s “consensus” (the regulative ideal of validity-claim redemption) are not synonyms but distinct moments of intersubjectivity, and both have specific work to do in describing what happens when a pair of preservice teachers writes a proof together. The paper closes with five discussion prompts designed to bootstrap dyadic pairs of preservice teachers through the level transitions, each tied to a Hegelian shape and a recognitive operation. The alignment is explicitly partial: Hegel’s *Phenomenology* covers shapes (Stoicism, Skepticism, Unhappy Consciousness, Spirit, Religion, Absolute Knowing) for which there is no van Hiele analogue, and van Hiele’s level theory has pedagogical structure

(the within-level “phases”) for which there is no direct Hegelian equivalent. We are mapping a structural family resemblance, not a bijection.

1. Introduction

The pedagogical literature on inquiry-based geometry for preservice teachers (PSTs) has converged on a small set of empirical findings and a much larger set of theoretical fragments. The empirical findings: PSTs entering teacher-preparation programs typically reason geometrically below the van Hiele level required of the children they will teach (Mayberry, 1983, *JRME*; Hourigan & Leavy, 2017, *The Teacher Educator*; Larbi, 2021; Armah, Cofie, & Okpoti, 2017; Armah, 2024; Salifu, 2018), with national variation but a robust pattern. Interventions that frame instruction explicitly through the van Hiele model can move PSTs forward (Yi, Flores, & Wang, 2020, *Teaching and Teacher Education*; Mbusi & Luneta, 2021; Mbusi, 2023; Shmigirilova et al., 2025; Vieira et al., 2023). The theoretical fragments are scattered. Some authors trace the developmental engine to *reflection* (Goffree, 1993, *ESM*, on Freudenthal); some to *abstraction* across “levels of proof” (Van Dormolen, 1977, *ESM*; Almeida, 2003, *IJMEST*); some to *commognitive* discourse participation (Wang, 2014; Wang, 2016; Mahlaba & Mahlobo, 2022, *Pythagoras*); some to “social climates” maintained by teachers (Bowers, Brandt, Stovall, & Vargas in *JRME*); some to misconception-tracking *phase-based instruction* (Mbusi & Luneta, 2021; Mbusi, 2023; Kandaga, Sumarmo, & Hendriana, 2022). What is missing is an account of *why* discourse, social climate, and reflection should be the mechanism — an account of the structural feature of geometric reasoning that requires intersubjective traffic to advance.

The argument of this paper is that the missing account is recognitive in the technical sense developed by Hegel in the *Phenomenology of Spirit* (1807/1977) and in two distinct contemporary readings: Brandom’s pragmatic, scorekeeping-based reading (Brandom, 1994, *Making It Explicit*; Brandom, 2019, *A Spirit of Trust*) and Habermas’s discourse-theoretic reading (Habermas, 1984; cf. the Brandom–Habermas exchange in Habermas, 2000, and Brandom’s reply in *European Journal of Philosophy*). The *Phenomenology*’s sequence of “shapes of consciousness” — Sense-Certainty, Perception, Force and Understanding, Self-Consciousness, Reason — tracks the same structural movement that the van Hiele levels track in the geometric domain: an object that consciousness takes itself to be tracking turns out, on examination, to require progressively more *mediation*, until consciousness recognizes that what it has been tracking all along is the structure of its own conceptual activity, intelligible only in the company of other knowers. The structural family resemblance between the two ladders is not coincidental, but it is also not a bijection. Hegel’s path includes shapes (Stoicism, Skepticism, the Unhappy Consciousness; Spirit; Religion; Absolute Knowing) for which there is no van Hiele analogue, and van Hiele’s level theory contains pedagogical structure (the within-level *phases* of information, guided orientation, explicitation, free orientation, and integration; van Hiele, 1986) for which there is no direct Hegelian counterpart. The alignment we develop is

structural and partial.

The pragmatic stake of the alignment is the design of dyadic discussion prompts for inquiry-based geometry courses for preservice teachers. PSTs in these courses typically operate at or near the upper van Hiele levels for many topics — though, as the empirical literature warns, with substantial variation across topics and populations — but they do not, in general, possess explicit theoretical self-understanding of what they are doing when they reason geometrically with a partner. A Hegelian scholar would name what they are doing (recognition, scorekeeping, the recognitive structure of the I and the We); the PST has the practice but not yet the vocabulary. The reciprocity claim runs in both directions: the theory gives the PST a vocabulary for what their classroom practice already accomplishes, which then becomes pedagogically reproducible with the children they will teach; the PST’s classroom practice gives Hegelian and post-Hegelian theory a concrete embodied site from which to work. Designing prompts that surface this reciprocity is the second contribution of the paper.

2. The Two Ladders, in Parallel

2.1 The first three shapes of consciousness in the *Phenomenology of Spirit*

Hegel’s *Phenomenology of Spirit* (Hegel, 1807/1977; cited by paragraph number per the Miller translation) opens with Sense-Certainty, the shape of consciousness that takes its object to be the immediate “This, Here, Now” (§90–110). Sense-Certainty cannot maintain its hold on its object: when consciousness tries to point out the immediate particular, “the pointing-out shows itself to be not an immediate knowing [of the point], but a movement from the Here that is meant through many Heres into the universal Here which is a simple plurality of Heres” (§108). The next shape, Perception, takes the object as a *Thing with many properties*, a One that must be reconciled with an “Also” of independent qualities (§111–131). Perception breaks down on its own contradictions: the Thing cannot simultaneously be a self-identical One and a collection of mutually external “matters” (§121, §129). The third shape, Force and Understanding (§132–165), posits a hidden inner world of laws standing behind the appearances and finds, in the course of trying to specify these laws, that “this absolute unrest of pure self-movement... is in the inner world that it has first freely and clearly shown itself” (§163) — that is, the structure consciousness has been describing as inner-world-of-laws is the structure of its own conceptual activity. At §164 the doorway opens: “consciousness of an ‘other,’ of an object in general, is itself necessarily self-consciousness.”

Hegel cautions, in the *Science of Logic*, that this is not the order of *logical* exposition. There he insists that the proper logical beginning of geometry is the most abstract element: “In geometry, the beginning has to be made not with a concrete space configuration but with point and line and then with plane figures, and among the latter not with polygons but with the triangle, and among

curves with the circle” (Hegel, *Science of Logic*, p. 714). This is the inverse of the van Hiele beginning. The *Phenomenology* gives the order in which a learning consciousness traverses a system; the *Logic* gives the order of conceptual exposition once the system is built. Geometric pedagogy is phenomenological, not logical, and this is the sense in which Hegel and van Hiele are talking about the same thing from different ends.

2.2 Self-Consciousness and Reason

The fourth shape, Self-Consciousness (§166–230), discovers that certainty of self requires *another* self-consciousness. The pivot is at §178: “Self-consciousness exists in and for itself when, and by the fact that, it so exists for another; that is, it exists *only in being acknowledged*.” Recognition is achieved “only when each is for the other what the other is for it” (§186). The Lordship and Bondage section (§190–196) demonstrates the pathology of asymmetrical recognition: the Lord, who claims authority without correlative responsibility, ends with a “truth” that is “the unessential consciousness and its unessential action” (§192). Symmetric recognition is the only stable form. It is in self-consciousness that “consciousness first finds its turning-point... and steps out into the spiritual daylight of the present” (§177), in the form Hegel names “‘I’ that is ‘We’ and ‘We’ that is ‘I.’”

The fifth shape, Reason (§231–359), takes the rational order of the world as its object. Observing Reason fails (the limit case is the bone-as-self-consciousness judgment of phrenology, §345) and gives way to active rational self-consciousness, which “must demand and produce its reality in an ‘other’” (§348) and emerges as Spirit and ethical life (§349). Beyond Reason, the *Phenomenology* continues through Spirit, Religion, and Absolute Knowing — shapes that have no van Hiele analogue and which are not part of the alignment we propose.

2.3 The van Hiele levels

The van Hiele levels of geometric thought (van Hiele, 1957/1985; van Hiele, 1986; van Hiele-Geldof, 1957/1985) are typically rendered as five: Visualization (Level 0 or 1, depending on numbering convention), Analysis, Informal Deduction (also called Abstraction or Ordering), Formal Deduction, and Rigor. Visualization treats the figure as a global gestalt; Analysis treats it as a bearer of properties; Informal Deduction begins to relate properties hierarchically (a square is a special rhombus, which is a special parallelogram); Formal Deduction works inside an axiomatized system; Rigor compares axiomatic systems. Within each level, van Hiele specified five *phases* of instruction (information, guided orientation, explication, free orientation, and integration) — internal pedagogical structure that does not map onto the Hegelian shapes and which we do not attempt to align.

The textbook canonical for inquiry-based geometry courses for preservice teachers in the U.S. (Aichele & Wolfe’s *Geometric Structures*, in the corresponding NotebookLM source) does not enumerate the levels but locates its central proof

method (“Five-Step Reasoning”) at “the highest van Heile levels” (Aichele & Wolfe, Ch. 11). The textbook’s pedagogical commitments — peer-to-peer discussion, “student tries first,” and the instructor’s discussion-prompt repertoire (“Can you convince the rest of us that that makes sense?”, “Do you agree? Disagree?”, “What do others think about what Janine said?”, in the textbook’s instructor-support source) — encode the recognitive structure of the highest levels in classroom procedure. Tall (2007, *ZDM*) gives the standard developmental gloss: in van Hiele’s geometric development “objects are first perceived as whole *gestalts*, then broadly described, with language growing more subtle so that descriptions may be formulated as definitions suitable for deduction and proof.”

3. The Structural Alignment

We propose a partial structural alignment. The mapping is presented as a series of correspondences with the relevant Hegelian operation in each case.

Visualization ↔ Sense-Certainty. The figure is taken as a singular *this*, immediately present, prior to articulation. The van Hiele Level-0 child who insists that a square rotated 45° is no longer a square but “a diamond” is reporting Sense-Certainty’s truth honestly. Hegel’s geometric example at ¶108 *is* this rung: the spatial “Here” of the figure, taken as immediate, reveals itself under pointing as “a movement from the Here that is meant through many Heres into the universal Here which is a simple plurality of Heres.” The level is unstable for the same reason Sense-Certainty is unstable: what is taken as a singular given is already a universal in disguise. Empirical reports consistently find a substantial population of school-age learners (and a non-trivial proportion of PSTs) operating at this level for unfamiliar geometric topics (Tandililing et al., 2025; Salifu, 2018; Cesaria et al., 2021; Larbi, 2021).

Analysis ↔ Perception. The figure is now a Thing with properties. Properties are real but unordered; the percipient holds them as an “Also” rather than as an articulated structure. The classic finding that a Level-1 student will not accept that a square is a special rectangle (because “a square has equal sides and a rectangle does not” treats “rectangle” as a determinate exclusive predicate rather than a higher-level universal) is exactly Hegel’s diagnosis of Perception (¶121, ¶129): the Thing is taken as a self-identical One that *cannot* be subsumed under more than one universal at once. Gutiérrez (2009, *RME* book review) glosses the matching pair: Level-1 reasoning is “very dependent on physical (mainly visual) characteristics of the examples shown,” while Level-2 reasoning “implies... the recognition of geometrical properties (i.e. ideal properties) as the main source of knowledge.”

Informal Deduction ↔ Force and Understanding. The student begins to treat the figure as the locus of *lawful relations*: properties imply properties; a square is a rhombus-with-right-angles is a rectangle-with-equal-sides. Hegel’s Force and Understanding posits an inner world of laws standing behind the play of appearance (¶143, ¶146), and at ¶163 discovers that the inner world of

laws is the structure of consciousness’s own conceptual activity. The geometric analogue is the moment when the implication-network is recognized as something *the conceptual scheme imposes on the figures*, not something the figures already contain. This is also the moment at which Aichele and Wolfe locate the textbook’s “Five-Step Reasoning”: justifying “how many different, seemingly unrelated, properties can be logical consequences of a simple definition” (Aichele & Wolfe, Ch. 11). It is the doorway to Self-Consciousness in Hegel’s terms because, having recognized that the inner-world-of-laws is the projection of consciousness’s own activity, consciousness has nowhere left to go for its truth except to other knowers of the same kind.

Formal Deduction $\leftrightarrow$ **Self-Consciousness.** This is the load-bearing transition, and it is where the recognitive engine (§4) does its main work. A formal proof is not the private psychological event of *seeing* that a conclusion must hold. It is a structured artifact submitted to other knowers for acknowledgment. Hegel’s claim at ¶178 — “it exists only in being acknowledged” — and at ¶186 — “only when each is for the other what the other is for it” — is not just structurally analogous to formal proof; it specifies what it *means* for a proof to be a proof. The textbook is unambiguous that “writing good Five-Step Reasoning arguments is a *communication skill*” (Aichele & Wolfe, Ch. 11). Empirical studies of PSTs at Level 3 confirm that the discursive structure is constitutive: Wang (2014, *RME*) analyzed two PSTs’ discourse on parallelograms at van Hiele Level 3 and found that “a single van Hiele level of thinking encompasses a range of complexity of reasoning and differences in discourse”; Mahlaba and Mahlobo (2022, *Pythagoras*) argued that “progress in the Van Hiele levels of geometrical thinking depends mainly on the discourse participation of the preservice teachers when solving geometry problems,” with “an explorative discourse... required for the development.”

Rigor $\leftrightarrow$ **Reason.** The student treats axiomatic systems themselves as objects, comparing Euclidean and non-Euclidean geometries, recognizing that the choice of axioms is itself a rational, communal practice. This is Hegelian Reason at the moment it realizes that “rationality is not an inert object to be observed; it is a practical, social reality that must be built” (¶347). Empirical work characterizing Level 5 (Arnal-Bailera & Guillén, 2023, *IJSME*; cf. Gutiérrez, Jaime, & Fortuny, 1991) used Delphi methodology — that is, *iterated expert consensus* — to validate the Level 5 indicators, an instructive (if accidental) reproduction of the recognitive structure at the meta-level: Rigor’s truth is itself stabilized by recognition among competent peers.

Where Hegel exceeds van Hiele

Hegel’s *Phenomenology* continues past Reason through Spirit (the social-historical-ethical realm), Religion, and Absolute Knowing, and includes within Self-Consciousness the further shapes of Stoicism, Skepticism, and the Unhappy Consciousness — the soul’s various attempts to live the lesson of Lordship and Bondage. None of these have direct van Hiele analogues, and we do not propose

to extend the alignment to them. The alignment is partial. Hegel is wider than van Hiele.

Where van Hiele exceeds Hegel

Van Hiele’s level theory is accompanied by a *phase* theory specifying the within-level pedagogical sequence (information, guided orientation, explication, free orientation, integration; van Hiele, 1986). The phases are pedagogical scaffolding internal to a level and do not correspond to anything in Hegel. Van Hiele’s empirical work is also explicit that level transitions are *language-mediated*, not purely cognitive — a claim that aligns with what Brandom and Habermas develop, but which van Hiele establishes on his own pedagogical grounds. The alignment we propose adds Hegelian theoretical depth to van Hiele’s structure without overwriting the parts of van Hiele’s theory that have no Hegelian counterpart.

4. The Recognitive Engine: Brandom’s “Trust” and Habermas’s “Consensus” Are Not Synonyms

The alignment in §3 is structural; it does not yet specify the *mechanism* by which a learner moves from one level to the next. Goffree (1993) glossed the mechanism as “reflection.” Bowers et al. (in *JRME*) glossed it as the “social climates” teachers create. Mahlaba and Mahlobo (2022) glossed it as “explorative discourse.” None of these is wrong; none specifies what makes discourse the operative mechanism. The specification we propose is that the mechanism is *recognition* in the technical sense developed in Hegel’s Self-Consciousness chapter and refined in two distinct contemporary readings: Brandom’s pragmatic reading and Habermas’s discourse-theoretic reading. Brandom’s “trust” and Habermas’s “consensus” are sometimes conflated in the secondary literature; they are not synonyms. They are different moments of intersubjectivity, and both have specific work to do in describing what happens when preservice teachers reason geometrically together.

4.1 Brandom: trust as deontic scorekeeping

Brandom (1994, *Making It Explicit*) models discursive practice as *deontic score-keeping*: each participant tracks their own and their interlocutors’ *commitments* (claims undertaken) and *entitlements* (the right to those claims) over the course of an exchange. To assert a claim is to undertake responsibility to vindicate one’s entitlement to that claim if challenged, and simultaneously to license others to inherit the claim from oneself. One mode of vindicating an entitlement is testimonial deferral: “B’s responsibility can be discharged by the invocation of A’s authority, upon which B exercised the right to rely. The buck is passed to A” (Brandom, 1994, p. 171). The discursive community emerges from the bottom up out of these dyadic transactions: “This is... an I-thou sociality rather than an I-we sociality. Its basic building block is the relation between an audience that is attributing commitments and thereby keeping score and a speaker who is

undertaking commitments” (Brandom, 1994, p. 508).

In *A Spirit of Trust* (Brandom, 2019), Brandom develops the recognitive structure further by reading the *Phenomenology*’s Self-Consciousness chapter through the scorekeeping vocabulary. Lordship and Bondage is recast as the failed asymmetrical recognition in which “the Master, who exercises authority (over the Servant) without correlative responsibility, is denominated the ‘independent consciousness’ and the Servant, who exhibits responsibility (to the Master) without correlative authority is denominated the ‘dependent consciousness’” (Brandom, 2019, ch. 9, p. 358 in MS pagination). The mature recognitive structure — the structural ideal that Hegel does not name but Brandom calls *trust* (Vertrauen) — is the reciprocal capacity to negotiate the inevitable disparity between commitments and their vindication through “confession and forgiveness” (Brandom, 2019, ch. 12). Crucially, Brandom rejects the consensus reading of Hegel: “It is only by ignoring or eliding the intricacy of that fine structure that one could think that the social dimension commits Hegel to a ‘consensus theory of truth,’ according to which objective truth becomes just whatever all community members agree on” (Brandom, 2019, p. 41).

Trust, on this picture, is a *practical capacity* exercised moment by moment in dyadic exchange. It does not require agreement. It requires the capacity to coordinate distinct perspectives over time without obliterating their distinctness. As Brandom replied to Habermas in their published exchange: “Conversational partners should not be pictured as marching in step, like soldiers on parade, but more as ballroom dancers, each making different movements... and thereby sharing a dance that is constituted precisely by the coordination of their individually different movements. Understanding... is a product of discursive coordination in which the distinctness of perspectives is maintained and managed. What is ‘shared’ in such a process is in principle not specifiable except by reference to the various perspectives from which it can appear” (Brandom, in *European Journal of Philosophy* 8/3, p. 362).

4.2 Habermas: consensus as the regulative ideal of validity-claim redemption

Habermas’s *Theory of Communicative Action* (1984) treats validity claims (truth, rightness, sincerity) as redeemable through discourse free of coercion among competent peers. The regulative ideal of consensus is what makes a claim *count* as valid; the actual practice of discourse approximates the ideal. Reviewing Brandom’s *Making It Explicit*, Habermas (2000) argued that scorekeeping fails to capture this regulative dimension: “in the case of a strategic team game such as baseball, it is a matter of a calculated adjustment to the reactions of others and not of a consensual co-operation that can satisfy the requirements of social integration... This picture falls short of establishing that the participants can converge in their intersubjective recognition of the same validity claim and can share knowledge in the strict sense of the verb” (Habermas, in *European Journal of Philosophy* 8/3, p. 346). For Habermas, the *I–Thou* of dyadic scorekeeping is

insufficient on its own; what makes the dyad’s exchange a knowledge-conferring practice is the regulative orientation toward consensus among all competent peers.

4.3 Carspecken’s blend: trust as practice, consensus as telos

Carspecken (in *Critical Ethnography in Educational Research* and in his subsequent work on inference fields) treats the two theorists as compatible at a higher level of abstraction while preserving their distinct operative roles: Brandom’s scorekeeping is the *mechanics* of illocution, Habermas’s consensus is the *telos* that makes the mechanics count as knowledge-conferring. In *Content Inference Fields in Intersubjective Space*, Carspecken writes: “Illocutionary claims are discussed by Robert Brandom (1998) in terms of commitments and entitlements that people in interaction keep track of on a ‘deontic scoreboard’... Habermas goes further than Brandom by articulating a theme we explore here in more detail than Habermas and yet still only partially. This is the idea that human understanding involves a ‘moment of mutuality’ as he calls it” (Carspecken, *Content Inference Fields in Intersubjective Space*, pp. 201–202).

We adopt Carspecken’s distinction. In what follows, *trust* is reserved for the moment-by-moment Brandom-style recognitive practice, and *consensus* is reserved for the Habermasian regulative ideal. The two come apart in the geometry classroom: a pair of preservice teachers who cooperate to write a proof exercises trust in the Brandomian sense whether or not they reach agreement on every step, and the proof they produce counts as valid (when it does) because it could in principle be defended in conditions approximating Habermasian ideal-speech consensus. The dyad lives the trust; the proof’s standing redeems the consensus.

4.4 What this means for level transitions

On this account, the transition from Informal Deduction to Formal Deduction is not a private cognitive achievement; it is constitutively a recognitive achievement, and the recognitive operation has two distinct moments. (i) The dyad has to develop the practice of *trust* — to make and accept inferential commitments, to defer entitlement to a partner’s testimony, to track the scoreboard of who is committed to what and on what grounds, to forgive partial errors and re-articulate. (ii) The proof the dyad produces has to be redeemable in *consensus* — it has to be the kind of artifact that any competent reader, not just the partner, could in principle recognize. A pedagogy that does not stage both moments will not stage the transition.

This specification supplies what was missing in the preceding theoretical fragments: it explains why “social climate” (Bowers et al., in *JRME*) is the mechanism — it is the institutional condition under which trust can be exercised; why “explorative discourse” (Mahlabla & Mahlobo, 2022) is the mechanism — it is the discursive register in which scorekeeping is visible; why “reflection” (Goffree, 1993) is the mechanism — reflection is the integration of the trust exercised

with one’s partner into an artifact redeemable in consensus. These are not three different mechanisms. They are three angles on the same recognitive structure.

5. State of the Literature

A comprehensive sweep of the major mathematics-education journals (JRME, ESM, ZDM, JMB, JMTE, IJSME, IJMEST, MTL, RME, MERJ; ~2,166 articles) and a Consensus search across the broader peer-reviewed literature for the period since 2018 confirm that the alignment proposed here has not been articulated directly. The closest extant work clusters into five groups.

(i) Hegelian framings of mathematics learning, not tied to van Hiele. Radford’s “objectification” framework is explicitly Hegelian. LaCroix (2014, *ESM*) summarizes: “Drawing on Hegelian phenomenology, objectification refers to a dialectical process of becoming consciously aware of, making sense of, and becoming critically conversant with cultural objects and systems of thought as one comes to participate within and contribute to cultural forms of activity, such as mathematics” (LaCroix, 2014, in *ESM* notebook source a2ed5fbd). Radford (2008, *ZDM*) cites Hegel’s *Phenomenology* in his discussion of semiotics and the objectification of algebraic knowledge. Baldino and Cabral (2006, *IJSME*) read inclusion and diversity in mathematics through a Hegel/Lacan lens. Stouraitis, Potari, and Skott (2017, *ESM*) examine “Contradictions, dialectical oppositions and shifts in teaching mathematics” within Activity Theory but do not connect to van Hiele. Ernest (2016, *ESM*) notes Lakatos’s *Proofs and Refutations* is “loosely based on Hegel’s Dialectic” but does not extend to van Hiele.

(ii) Brandomian framings, not tied to van Hiele. Bakker, Smit, and Wegerif (2015, *ZDM*) cite Brandom’s *Articulating Reasons* on dialogic teaching and scaffolding. Sensevy, Forest, Quilio, and Morales (2013, *ZDM*) cite Brandom’s *Making It Explicit* in their cooperative-engineering framework. Rüede (2013, *ESM*) cites Brandom in his analysis of how secondary students structure algebraic expressions. None apply Brandom to geometric learning or to van Hiele transitions.

(iii) Habermasian framings, not tied to van Hiele. Ongstad (2006, *ESM*) develops a triadic semiotic framework for mathematics communication based extensively on Habermas’s communicative action: “with a speech act ‘MP’, S takes up relations simultaneously to something in the objective world, to something in the subjective world, and to something shared in a social world” (Ongstad, 2006, in *ESM* notebook source 09d3f35e). Díez-Palomar and Olivé (2015, *ZDM*) cite Habermas’s *Theory of Communicative Action* in their study of interactive groups and dialogic talk. Morselli and Boero (in *Early algebraization*, reviewed in *RME*) interpret students’ algebraic proving through Habermas’s theory of rational behavior. None connect to van Hiele.

(iv) Discourse-theoretic readings of van Hiele, not Hegelian. Wang’s (2014, 2016) commognitive analysis of geometric discourse at van Hiele Level 3 and Mahlaba and Mahlobo’s (2022) commognition $\times$ van Hiele theoretical review are

the closest extant work to the present argument. Both use Sfard’s commognitive framework rather than Brandom or Hegel. Both identify discourse participation as the level-transition mechanism. Wang (2014, *RME*) explicitly observes that “a single van Hiele level of thinking encompasses a range of complexity of reasoning and differences in discourse and thus a deeper investigation of students’ mathematical thinking within assigned van Hiele levels is warranted” (Wang, Sasha, 2014, “How do they know it is a parallelogram?”, *RME*).

(v) Empirical work on van Hiele in PST education. A substantial body of empirical literature characterizes PST geometric thinking and tests interventions (Mayberry, 1983, *JRME*; Burger & Shaughnessy, 1986, *JRME*; Gutiérrez, Jaime, & Fortuny, 1991, *JRME*; Mayberry’s 19 PSTs from 1983 remain widely cited at 239 citations; Hourigan & Leavy, 2017; Yi, Flores, & Wang, 2020; Larbi, 2021; Armah et al., 2017, 2024; Mbusi, 2021, 2023; Vieira et al., 2023; Shmigirilova et al., 2025; Tao et al., 2024). Bjuland (2004, *ESM*) studied “student teachers’ reflections on their learning process through collaborative problem solving in geometry” — a direct precursor to the dyadic-pair design proposed in §6. Komatsu, Tsujiyama, and Sakamaki (2014, *IJMEST*) document a paired interaction in which two students jointly reflect on a proof to handle a counterexample, exhibiting in transcript what we describe theoretically as the trust-moment of recognitive practice. Appova and Taylor (2017, *JMTE*) report that expert mathematics teacher educators use the van Hiele framework explicitly to “bridge theory and practice” with PSTs through sorting tasks and the analysis of K-8 students’ work — the same theory–practice reciprocity we identify in §6.

The synthesis of (i)–(v) is straightforward: the materials for the alignment proposed in this paper exist in the literature, scattered across journal traditions and theoretical vocabularies, and the alignment itself has not been written.

6. Discussion Prompts to Bootstrap Dyadic Pairs Through the Levels

The pedagogical claim of this paper is that the level transitions can be staged in *dyadic discussion sequences* whose structure mirrors the recognitive operations described in §4. Each prompt below pairs a geometric scenario with a dyadic-interaction protocol and a brief instructor gloss naming the Hegelian shape and the recognitive operation. The prompts are designed for preservice teachers, who in this kind of inquiry course typically operate at or near the upper van Hiele levels for many topics (with empirical caveats from Mayberry, 1983; Hourigan & Leavy, 2017; Larbi, 2021; Armah, 2024) but who do not, in general, possess explicit theoretical self-understanding of what they are doing. The prompts are designed so that the practice happens first and the theory becomes available afterward, by retrospective recognition of what the dyad just did.

The instructor gloss is intended for the instructor’s own framing, not for the prompt-as-given to students. The student version of each prompt is the geometric

scenario plus the dyadic protocol. Whether to disclose the theoretical structure to students is a separate pedagogical decision, contingent on the course’s epistemic register.

Prompt 1 — Visualization → Analysis (Sense-Certainty → Perception). The rotated square. *Scenario:* A child in your future classroom holds up a square cut from paper, then rotates it 45° . The child reports that it is now “a diamond, not a square.” The textbook will say the child is wrong: rotation does not change the figure’s identity. The child reports honestly what they see. *Protocol:* Each partner writes their response to the child privately for five minutes. Partners exchange notes. The dyad has 15 minutes to find a *form of words* (one or two sentences) they can both stand behind that they would say to the child. Disagreement is fine; the goal is the form of words, not agreement. *Instructor gloss:* This is the Sense-Certainty/Perception transition. The pointing-out of “this square” collapses into “a movement from the Here that is meant through many Heres” once orientation enters the analysis. The dyad’s joint form of words is their first articulated universal — their first recognition that the figure is a Thing-with-properties, one of which is *rotation-invariant identity*.

Prompt 2 — Analysis → Informal Deduction (Perception → Understanding). Properties as a list, properties as a structure. *Scenario:* A parallelogram is drawn on the board. Each partner has five minutes to list, privately, every property of a parallelogram they can produce. *Protocol:* Partners exchange lists. For each pairing of properties from the two lists, they must classify (a) “alternative wordings of the same property,” (b) “separate properties, both true,” or (c) “one implies the other; which way?” When they reach a pair they cannot classify, they have located the spot at which Perception is no longer adequate to the figure. *Instructor gloss:* This is the Perception/Understanding transition. The properties are real (Perception’s truth), but they are not yet ordered into a lawful structure (Perception’s failure). The classification task forces the dyad to begin treating the figure as the locus of *implication relations* — Force and Understanding’s hidden inner world. The “spot they cannot classify” is the doorway: the moment they realize that whatever they decide will be *imposed by their conceptual scheme* rather than read off the figure.

Prompt 3 — Informal Deduction → Formal Deduction (Understanding → Self-Consciousness, Brandomian trust). *Scenario:* The textbook claims “the diagonals of a kite are perpendicular.” *Protocol:* Each partner privately writes a Five-Step-Reasoning proof of the claim (15 minutes). Partners exchange proofs. The receiving partner does *not* grade, agree, or correct. Their job is to register, on a marked-up version of the proof, exactly *which step they are willing to inherit on the writer’s authority* and *which step they would have to verify themselves before passing it on to a third party*. They return the marked-up proof to the writer. The dyad then discusses, for ten minutes, the steps each was willing to inherit and the steps each was not. *Instructor gloss:* This is the Brandomian trust-moment, made visible. “Inheriting on the writer’s authority”

is exactly testimonial deferral in Brandom’s sense (1994, p. 171). Marking the steps the receiver would not inherit is making the deontic scoreboard visible. The dyad’s discussion is the recognitive synthesis: each is becoming, in Hegel’s phrase, “for the other what the other is for it” (§186). The transition into Formal Deduction *is* the practice of mutual scorekeeping. A proof that no partner is willing to inherit is not a proof; a proof every step of which is inherited only on testimony is also not a proof. The proof emerges in the discrimination.

Prompt 4 — Lordship and Bondage / asymmetric recognition resolved into reciprocal recognition. *Scenario:* A variation of Prompt 3 in which the partners disagree, after discussion, about whether a particular step is justified. *Protocol:* Neither partner may proceed by overruling the other or by deferring to the instructor. The dyad has 20 minutes to either (a) arrive at a re-articulation of the disputed step that both can either accept or reject for the same reason, or (b) construct a counter-figure (or a counterexample, or a missing case) that determines the question. The instructor circulates but does not adjudicate. If neither resolution is reached in 20 minutes, the dyad writes a paragraph naming what *would* count as a resolution. *Instructor gloss:* This is the Lordship/Bondage moment. A dyad that resolves the disagreement by one partner overruling the other has produced asymmetric recognition: the “winner” has authority without correlative responsibility, and Hegel’s diagnosis (§192) applies — the truth so produced is “the unessential consciousness and its unessential action.” A dyad that re-articulates *or* that produces a counter-figure has produced reciprocal recognition: each is for the other what the other is for it (§186). A dyad that writes the paragraph naming the conditions of resolution is doing what Hegel does in the *Phenomenology* itself — making explicit the structure that would let the recognition succeed. All three outcomes are valuable; the instructor’s role is to make the difference legible afterward.

Prompt 5 — Formal Deduction → Rigor (Self-Consciousness → Reason, Habermasian consensus). Two definitions of a kite. *Scenario:* The textbook gives two non-identical-looking definitions of a kite (e.g., “a quadrilateral with two pairs of adjacent equal sides” and “a quadrilateral with at least one diagonal of symmetry”) and proves they are equivalent. Each partner is given one of the two definitions privately and is told only that the other partner has a different (also-correct) definition. *Protocol:* Without disclosing their definition to each other, each partner derives three properties of a kite from their definition. They exchange property lists. They then together attempt to determine: do the property lists describe the *same figure*? If yes, can they reconstruct what the other partner’s definition must be? If no, can they construct a figure that satisfies one definition but not the other? *Instructor gloss:* This is the move into Rigor and the Habermasian regulative dimension. The dyad’s task forces them to step *outside* the system (working with one definition) into *across-system* reasoning (comparing two definitions of the same object). In Hegel’s vocabulary, the move is from working inside Reason’s first pass (Observing Reason) to actively producing rationality “through its own activity” (§347). In Habermas’s vocabulary, the question “do these property lists describe the same figure?” is

precisely the question of whether the validity claims are convergent — whether two distinct dyadic-trust practices have produced an artifact redeemable in the imagined community of all competent geometers. The dyad is now doing meta-mathematical work: they are not proving facts about kites; they are recognizing the conditions under which two definitions can claim to be definitions of *the same* kite. This is what teachers of geometry have to do silently every time they decide which definition of a polygon to teach.

Theory–practice reciprocity

The reciprocity claim is not that teaching PSTs Hegel will improve their geometry, nor that teaching them van Hiele will improve their philosophy. It is more specific. PSTs in inquiry-based geometry courses already perform recognitive operations of the kind described in §§2–4. They acknowledge each other’s commitments, defer to each other’s authority, register where they cannot inherit a partner’s claim, and forgive partial errors as they re-articulate together. They do this without theoretical vocabulary for it. A Hegelian-Brandomian framing of what they are doing makes the recognitive structure newly visible to *them* — and what is newly visible to them as PSTs is what they will be able to design *for* the children they will teach. The pattern documented by Appova and Taylor (2017, *JMTE*), in which expert MTEs use van Hiele explicitly to bridge theory and practice with PSTs, generalizes: van Hiele names the developmental structure of the geometric practice; Hegel via Brandom names the recognitive structure that drives transitions through the developmental structure; the dyadic prompt is the classroom site at which both come into the PST’s experience together.

7. Conclusion

A partial structural alignment between the *Phenomenology of Spirit*’s shapes of consciousness and the van Hiele levels of geometric thought lights up an under-specified mechanism for level transitions: the recognitive practice that Brandom calls *trust* (moment-by-moment dyadic scorekeeping) and the regulative ideal that Habermas calls *consensus* (validity-claim redemption in the conditions of an imagined community of competent peers). The two moments are not synonyms; they are not interchangeable; they have to be staged together in the geometry classroom for the level transitions to occur. Inquiry-based geometry courses for preservice teachers are an unusually well-suited site for this staging, because the textbook commitments and the discussion-based pedagogy already encode the recognitive structure in classroom procedure. The five dyadic prompts proposed here are first sketches; the next step is empirical, comparing dyadic interactions under these prompts to van Hiele level transitions in PST cohorts, with discourse coding informed by both the commognitive (Sfard, Wang, Mahlaba & Mahlobo) and the recognitive (Brandom, Carspecken) traditions. The alignment we have proposed between Hegel and van Hiele is partial, structural, and not a bijection. Hegel exceeds van Hiele in scope; van Hiele exceeds Hegel in pedagogical specificity. The alignment is useful to the extent that it specifies an engine that both

theories implicitly require but neither explicitly provides on its own.

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